



University
of Glasgow

W11-01: Answering Exam Questions

COMPSCI4064 (H) and COMPSCI5088 (M)

Dr. Richard McCreddie

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Exam Basics

- We are evaluating what you (should) have **learned from the course**
 - We only evaluate you on what we have **taught** (what is on or linked to from the Moodle)
 - Course assessment planning considers the entire set of topics covered and we cannot assess everything, this means if we previously assessed something via exercise, it very likely will not turn up during the exam
- Honors and Masters students have a slightly different exam
 - The Masters one should be a little harder (as masters is a higher qualification)
 - Both are **60-mark exams**
 - You have **1 hour 30 minutes** (Fixed time Exam)
 - (no additional submission time this year as its done via Moodle Quiz)
 - Marking is declarative, i.e. there are valid answers and the markers check whether your answer matches (fully or partially)



Question Format

- Questions are divided into three components
 - **Context:** This provides some background information about the scenario that the question should be answered in relation to
 - **Question:** The question you should directly answer.
 - **Guidance:** How the marks are awarded for the question.
- Context carries over between sub-questions of each major question
 - So the context for 1. (a) may be relevant for 1. (b), but not for 2. (a)

1. (a) **Context:** Consider the following scenario. You are a big data engineer working at a satellite imaging company, who are contracted to develop an alerting system for natural disasters based on the images captured by the satellites. When active, a satellite sends a high-resolution image to a base station on the ground each second, which is then streamed to the company headquarters to be processed. Each image is around 10 MB in size. The company owns 100 satellites with the goal of doubling that in the next 5 years. Satellites are set in fixed orbital paths and are tasked with taking pictures from specific regions on earth. When not over a target region the satellite is inactive (to save power). Typically, around 20% of the satellites are active at any one time, although this may increase to 50% during busy periods of the year (e.g. US Hurricane season).

Question: Processing the images produced by these satellites for disaster alerting constitutes a big data task. List the dimensions/properties of the task that makes it a big data problem and discuss why.

Guidance: The question is worth 6 marks. There are 3 marks for correctly identifying the dimensions/properties and 3 marks for the associated discussion (1 mark per correct point made).

[6]



Question Types

- Be aware of the **language** used in the '**Question**' block
 - **Discuss..., Describe...:** For these types of questions you are expected to write a short paragraph. A good rule of thumb is take the number of marks for the question and divide by 2, that is approximately the number of sentences to write.
 - **Lists...:** Other types of questions can usually be answered using a bullet-point list of some short, check the guidance statement to see if there is a clear 'X points are expected' statement, if so write one bullet per point.



Describe Example

- For describe type questions, we are looking for your answer to cover particular points
 - See how there are [X Mark] statements in the answer
 - The answer scripts are what the markers are given
- Markers are seeing if what you wrote matches closely enough to the correct answer to award the mark

(c) **Context:** One of the foundations of a decentralized solution like Cassandra is that it uses hashing schemes to enable fast indexing of its available storage.

Question: Describe consistent hashing and order-preserving hashing, and then discuss how they are combined in Cassandra.

Guidance: This question is worth 6 marks. 4 marks are available for explaining how the two hashing schemes function, and 2 marks are available for describing how they can be combined.

[6]

Solution: Both schemes are referred to as hashing schemes, but operate at different levels. Consistent hashing will partition the ID space across existing nodes, [1 mark] and will assign an ID X to the first node whose ID is larger than (or equal) to X . [1 mark] This applies regardless of how ' X ' was computed. An order-preserving hash function can then be used to map an input keys Y to an ID A on the ID space in an order-preserving manner [1 mark] (i.e., guaranteeing that any key $Z > Y$ will be mapped to an ID $B > A$), then consistent hashing would be used to map A to a node. [1 mark]. Combination discussion is worth [2 marks].



List Example

- If you see 'X... points are expected', the question is probably expecting a list answer
 - You can be shorter in the writing for these
 - Often there will be multiple marks per bullet
- The markers are usually looking for a near match against the answer sheet for these

3. (a) **Context:** A large shipping company is in the process of upgrading the storage infrastructure that their company uses to store data produced by the ships in their fleet. The company needs to store two types of data: 1) audio logs of communication between its ships and third parties (coast guards, docking facilities, etc.) that are relatively large files and 2) sensor data produced by the ships themselves (GPS coordinate data, engine temperature, etc.) that comprise many small files.

Question: The company currently uses an NFS backed by RAID6 storage, which is replicated to an off-site back-up at the end of each week and is considering moving to an HDFS-based solution. What are the advantages and disadvantages of such a move?

Guidance: This question is worth 8 marks. 4 distinct points are expected. For each point, one mark is available for what difference between the two solutions would be and a further mark is available for what impact this would have. You should connect these differences to Big Data aspects like resiliency, scalability, etc.

[8]

Solution:

We expect the student to compare and contrast the NFS+RAID solution with HDFS. Valid points for discussion include:

- Moving to an HDFS solution would provide replication of files on-site [1 Mark], enabling better throughput due to more disk and network capacity [1 Mark].
- Having a cluster of machines [1 Mark] protects the company against hardware failure in the NFS server (a single point of failure) [1 Mark].
- On the other hand, an HDFS is more complex to manage and requires orchestration [1 Mark] as multiple servers running data nodes are needed [1 Mark].
- An HDFS solution would also be more expensive [1 Mark], as more redundant hardware would be needed [1 Mark].
- HDFS Storage policies can be used to support intelligent selection of storage devices for files [1 Mark], which is not supported by the NFS [1 Mark].
- Data access speeds will become faster for audio log retrieval, but slower for sensor data [1 Mark], as the HDFS is optimised for large file IO [1 Mark].

Other answers are acceptable under the same paired mark format.



University
of Glasgow

Course Coordinator
Dr. Richard McCreadie

Email: richard.mccreadie@glasgow.ac.uk
Room: SAWB 304

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